THE DELTA-INVARIANT AT A POINT IS UNCHANGED BY EXTENSION OF THE BASE FIELD UNDER THE GEOMETRIC NORMALITY HYPOTHESES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites five cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

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Allowed standard prerequisites (editorial).

- Geometrically normal in codimension one: assumptions and consequences. Stacks Project, Varieties, Lemma lemma-geometrically-normal-in-codim-1 (tag 0C3T). Supplies geometric reducedness of A and normality of A' tensor_k K, which is what upgrades the inequality to an equality.
- The pre-delta construction. Stacks Project, Varieties, Lemma lemma-pre-delta-invariant (tag 09NL). Provides the intermediate ring A' whose base change is compared on both sides.

Source reference. Original lemma 0C3P: sources/stacks-chow/tags.txt, source lines 15199–15199. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0C3S: sources/stacks-chow/tags.txt, source lines 15202–15202. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0C3R: sources/stacks-chow/tags.txt, source lines 15201–15201. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02M1: sources/stacks-chow/tags.txt, source lines 3428–3428. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X be a locally algebraic k-scheme. Let K/k be a field extension and set $Y = X_K$. Let $y \in Y$ with image $x \in X$. Assume assumptions (a), (b), (c) of Lemma 0C3P hold for $x \in X$ and that $\dim(\mathcal{O}_{Y,y}) = 1$. Then the δ-invariant of X at x is δ-invariant of Y at y.*

Proof. Set $A = \mathcal{O}_{X,x}$ and $B = \mathcal{O}_{Y,y}$. By Lemma 0C3P we see that A is geometrically reduced. Hence B is a localization of $A \otimes_k K$. Let $A \to A'$ be as in Lemma 0C3S. By Lemma 0C3P we see that $A' \otimes_k K$ is normal. Hence

$$B' = B \otimes_{(A \otimes_k K)} (A' \otimes_k K)$$

is normal, finite over B, and $B \to B'$ induces an isomorphism on total rings of fractions. Namely, pick $f \in \mathfrak{m}_A$ satisfying (1) – (6) of Lemma 0C3R; since $\dim(B) =$

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1 we see that $f \in \mathfrak{m}_B$ playes the same role for B and we see that $B_f = B'_f$ because $A_f = A'_f$. It follows that $B \to B'$ is as in Lemma 0C3S for B. Thus we have to show that $\text{length}_A(A'/A) = \text{length}_B(B'/B) = \text{length}_B((A'/A) \otimes_A B)$. Since $A \to B$ is flat (as a localization of $A \to A \otimes_k K$) and since $\mathfrak{m}_B = \mathfrak{m}_A B$ (because $B/\mathfrak{m}_A B$ is zero dimensional by the remarks above and a localization of $K \otimes_k \kappa(x)$ which is reduced as $\kappa(x)$ is separable over k) we conclude by Algebra, Lemma 02M1. $\square$